

SAR Dust

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Abstract

We explore the impact of noise attacks on Synthetic Aperture Radar (SAR) image classification and gain a better understanding of the limits of robustness for CNN classifiers [1]. We benchmark ResNet-18 [2] classification performance against 9 different noise classes with varying noise intensity. We then explore how we can improve model robustness to noise. By segmenting a fraction of the SAR targets in the training set, we can improve classification with increased levels of noise which the model previously performed poorly on.

Data

All SAR figures shown are from the MSTAR dataset. The original dataset contained 14 classes with 1,000 samples per class, and we use a 70/20/10 train, validation, test split.

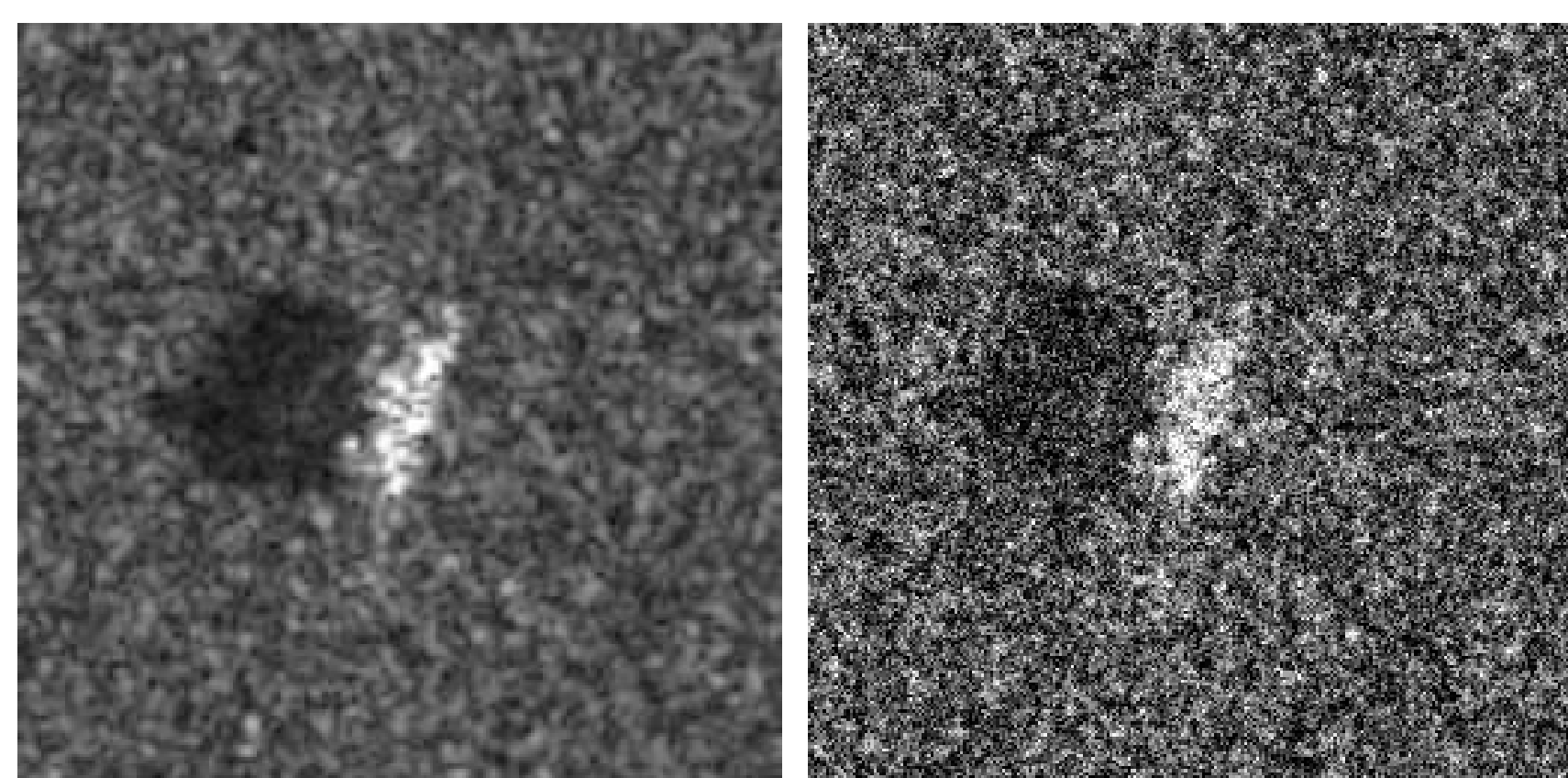


Figure 1: Original SAR Image (left) and Gaussian Noise Image with PSNR 15.84 dB (right)

Technical Approach

Benchmarking

- Implemented 9 noise models to replicate different types of interference in SAR imaging.
- Trained ResNet-18 backbone with clean SAR data, and tested with added noise.
- Metrics: Peak Signal to Noise Ratio (PSNR), overall and per-class accuracy, entropy, expected calibration error (ECE), t-SNE plots, and saliency maps.

Isolated Target Training

- Applied a Bilateral Speckle Filter [3] and calculated the mean, μ , and standard deviation, σ , of the smoothed pixel distribution to create an adaptive threshold $T = \mu + 2\sigma$.
- The isolated target samples are mixed 50/50 into the training set with original samples.
- We analyze the shape bias in our model by inspecting saliency maps produced via the Guided Grad-CAM algorithm [4].

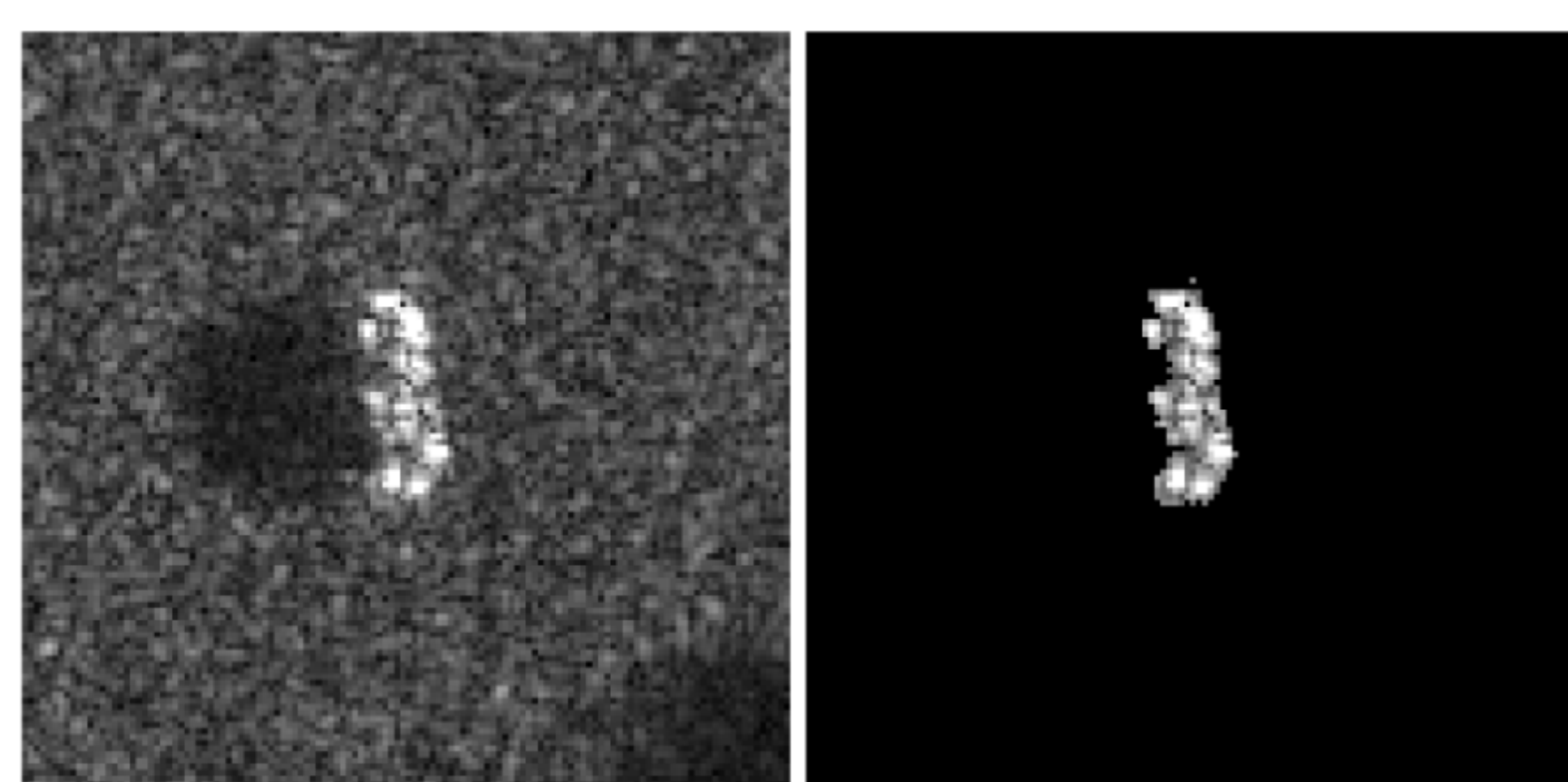


Figure 2: BTR60 Original Image (left) and Isolated Target Image (right)

Results

Benchmarking

- Model achieves 97.14% accuracy without added noise.
- The degradation shown in Figure 3 shows the model generalizes well to small amounts of noise, before collapsing on to a single class.
- Figure 4 shows the model collapses at PSNR of around 10-15 dB. This result was consistent across all noise models.

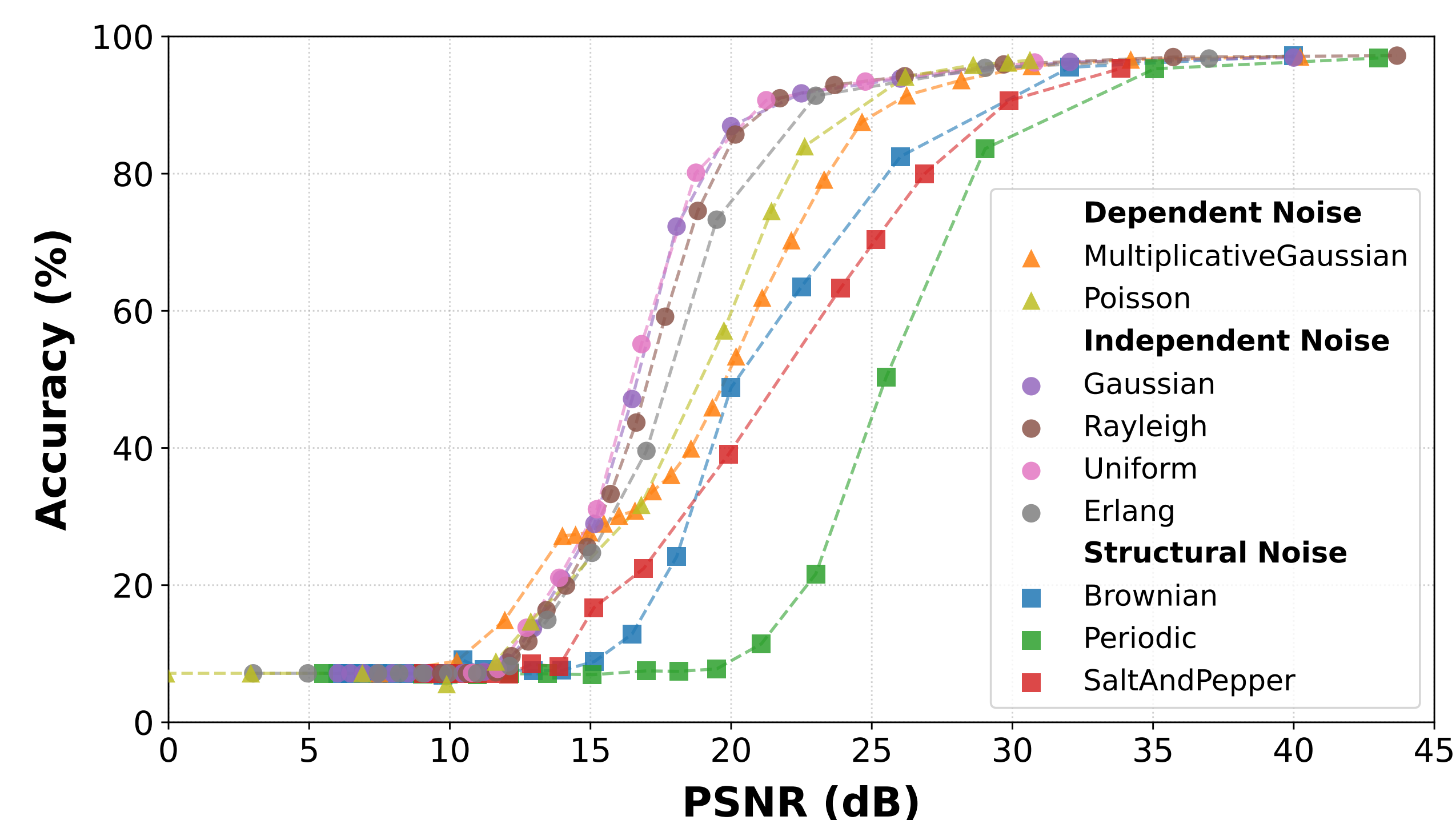


Figure 3: PSNR vs. Accuracy

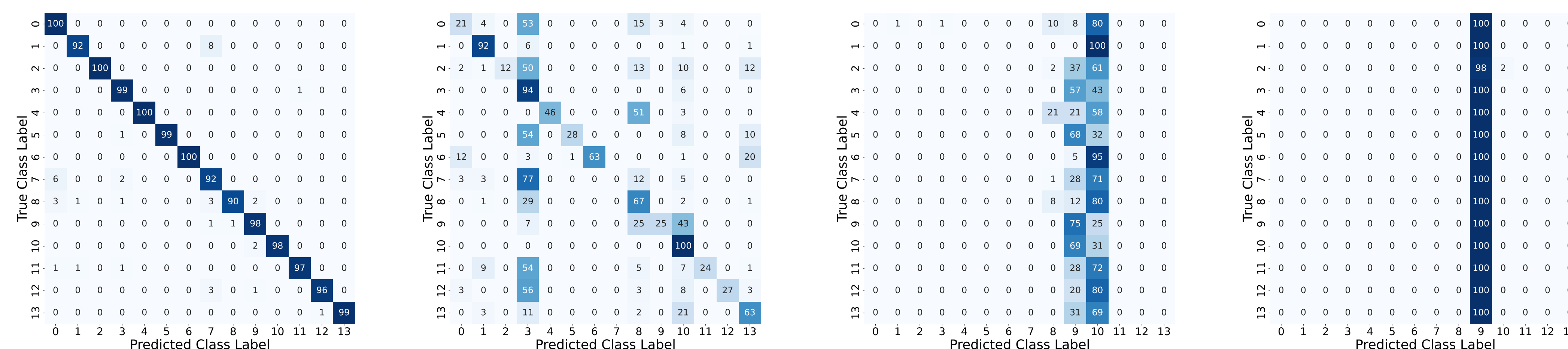


Figure 4: Confusion matrices: No Noise, Gaussian Noise with PSNR 16.48 dB , 12.04 dB , and 10.46 dB .

Isolated Target Training

- Our 50/50 model achieves an accuracy of 97.93% without added noise.
- Figure 5 shows our 50/50 model handles noise better than the original model, with a maximum improvement in accuracy of 66.7%.
- Figure 6 shows that the 50/50 model has increased shape bias, as more weight has moved off of the shadows and on to the target.

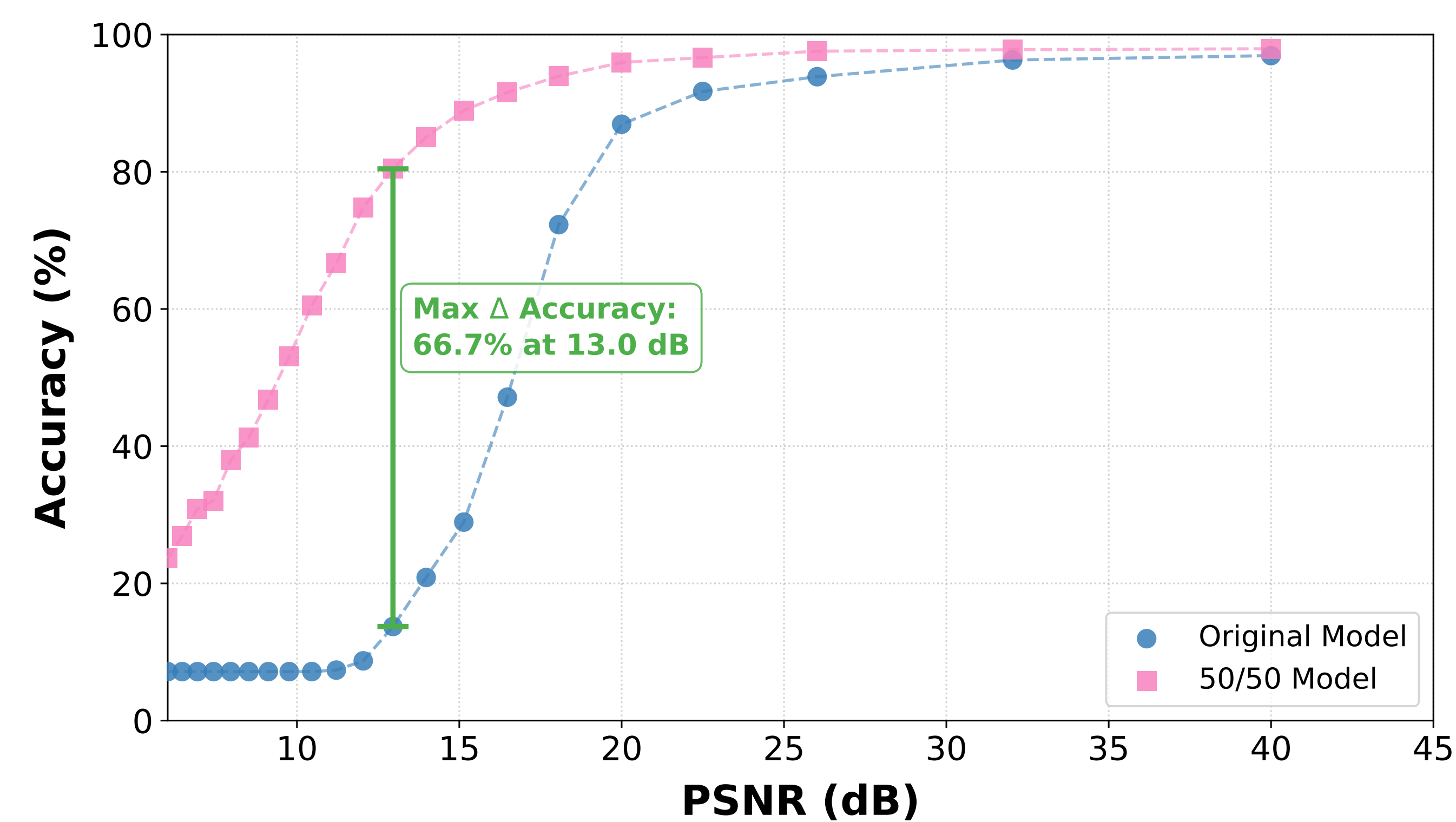


Figure 5: 50/50 Model vs. Original Model Performance (Gaussian Test)

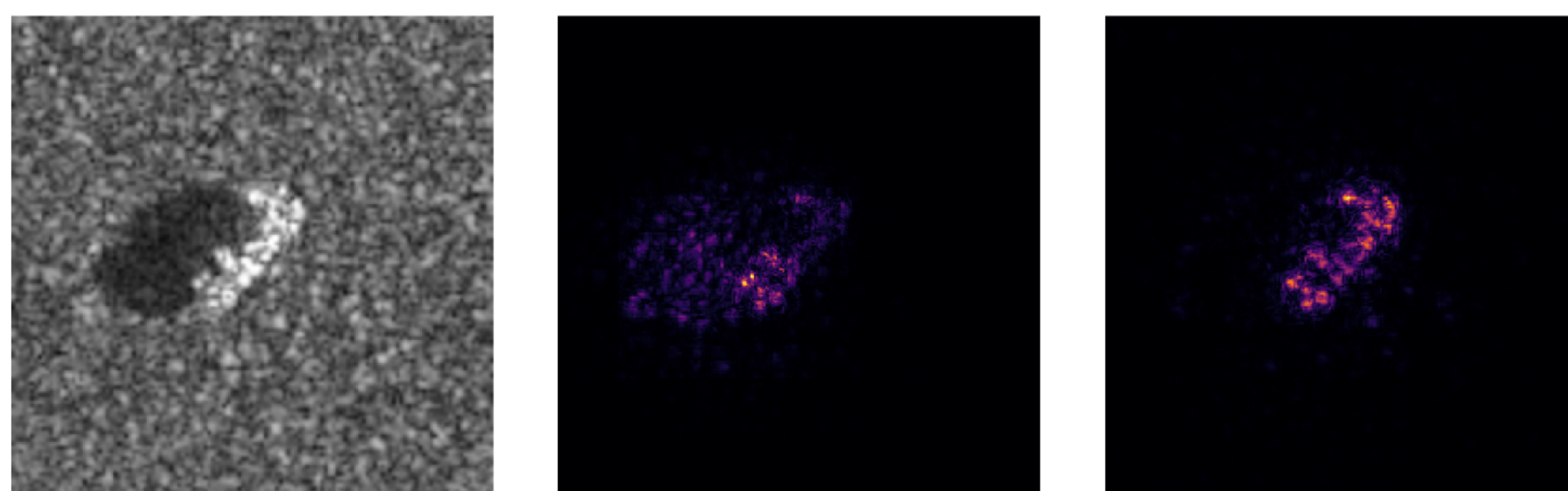


Figure 6: BTR70 Original Image (left), Original Model Saliency (middle), and 50/50 Model Saliency (right)

Conclusions

- We show that CNN's are able to classify SAR imagery with accuracy greater than 97%, but struggle to generalize to noisy data with PSNR less than 20 dB.
- We show that adding isolated targets to the training data induces shape bias, improving accuracy by 66.7% at PSNR of 13 dB.

Future Work

While our adaptive-thresholding provides an effective baseline, low-contrast targets are inherently harder to isolate. Replacing this rule-based thresholding with learnable modules in the network may provide even greater improvements to robustness.